

Assignment - 1

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Section: 2

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Initial

Course : PHY112

Code

Answer to the question no 1

Q)

Electric field produced by a charged ring -

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{qz}{(\sqrt{z^2 + R^2})^3}$$

Here,

q = charge on the ring

z = distance from the ring to the point P

R = the radius of the ring

$$\begin{aligned}\therefore E &= \frac{1}{4\pi\epsilon_0} \times \frac{10 \times 10^{-12} \times 3.5}{(\sqrt{(3.5)^2 + (0.15)^2})^3} \\ &= 9 \times 10^9 \times \frac{10 \times 10^{-12} \times 3.5}{(\sqrt{(3.5)^2 + (0.15)^2})^3} \\ &= 7.32 \times 10^{-3} \text{ N/C}\end{aligned}$$

$$\begin{aligned}q &= 10 \text{ pC} \\ &= 10 \times 10^{-12} \text{ C}\end{aligned}$$

$$z = 3.5 \text{ m}$$

$$\begin{aligned}R &= 15 \text{ cm} \\ &= 0.15 \text{ m}\end{aligned}$$

Therefore, the magnitude of the electric field produced by the ring at point P is

$$7.32 \times 10^{-3} \text{ N/C}$$

b)

Direction of the electric field is towards z axis. which is outwards direction.

$$\begin{aligned}\vec{E}_p &= \frac{1}{4\pi\epsilon_0} \times \frac{10 \times 10^{-12} \times 3.5}{(\sqrt{(3.5)^2 + (0.15)^2})^3} \hat{k} \\ &= 7.32 \times 10^{-3} \hat{k}\end{aligned}$$

c)

If we consider the same amount of charge (10 pC) spread uniformly in a disk of the same radius (15 cm), we can use the equation for the electric field produced by a charged disk.

$$E = \frac{\sigma}{2\epsilon_0} \times \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$$

Here,

σ = charge density

ϵ_0 = permittivity of free space

z = distance from the disk to the point P

R = the radius of the disk

We know,

$$\sigma = \frac{q}{A}$$

$$= \frac{q}{\pi R^2}$$

$$= \frac{10 \times 10^{-12}}{\pi \times (0.15)^2}$$

$$= 1.41 \times 10^{-10} \text{ C m}^{-2}$$

$$q = 10 \text{ pC} \\ = 10 \times 10^{-12} \text{ C}$$

$$R = 15 \text{ cm} \\ = 0.15 \text{ m}$$

$$\therefore E = \frac{1.41 \times 10^{-10}}{2\epsilon_0} \times \left(1 - \frac{2.5}{\sqrt{2.5^2 + 0.15^2}} \right)$$

$$z = 2.5 \text{ m}$$

$$= 0.014 \text{ N/C}$$

(Ans)

d)

To find the charge density of the infinite sheet, we need to take the limit of the charge density of the disks as the radius goes to infinity.

$$\lim (R \rightarrow \infty)$$

$$\therefore \sigma = \frac{q}{A}$$

$$= \frac{q}{\pi R^2}$$

$$= \frac{10 \times 10^{-12}}{\pi \times (\infty)^2}$$

$$= 0$$

2)

For an infinite plane sheet,

$$\vec{E}_p = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{R^2 + \left(\frac{z^2}{R^2} + 1\right)}} \right)$$

$$= \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{R} \right) \quad [\because z = 0]$$

$$= \frac{\sigma}{2\epsilon_0}$$

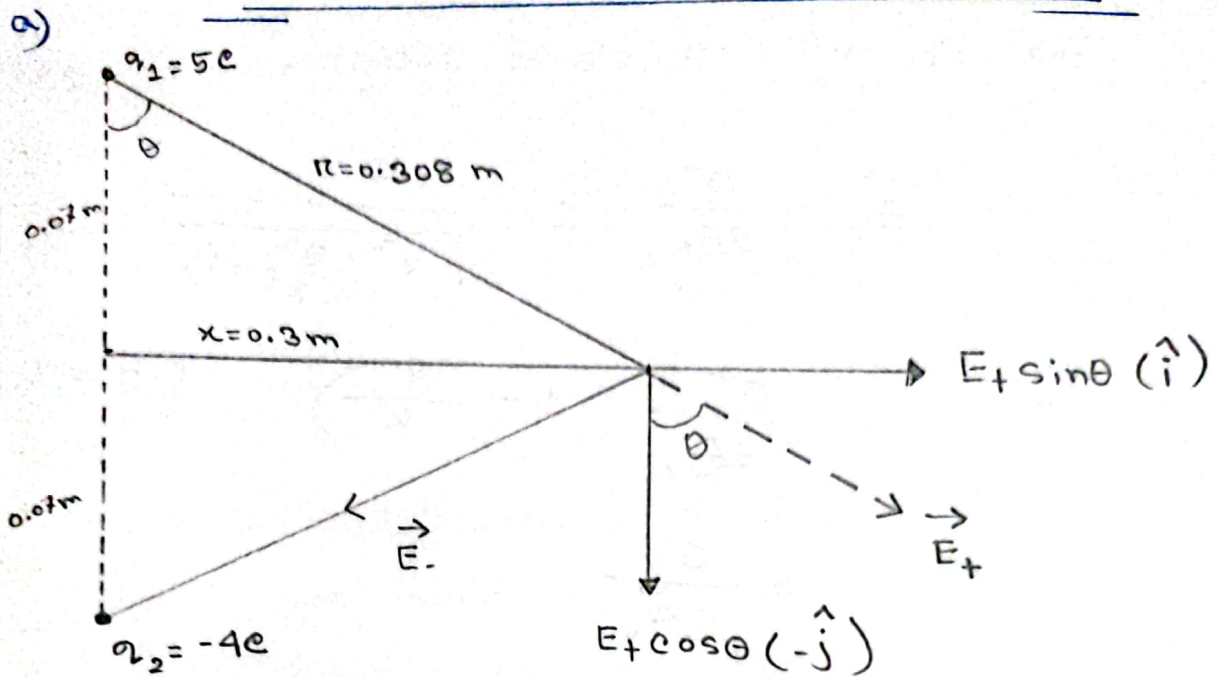
$$\therefore \sigma = 2\epsilon_0 E_p$$

$$= 2\epsilon_0 \times 0.014$$

$$= 2.48 \times 10^{-13} \text{ C/m}^2$$

(Ans)

Answer to the question no 2



$$\vec{E}_+ = E_+ \sin \theta \hat{i} + E_+ \cos \theta (-\hat{j})$$

$$\vec{E}_- = E_- \sin \theta (-\hat{i}) + E_- \cos \theta (-\hat{j})$$

Here,

$$\sin \theta = \frac{x}{\sqrt{r^2 + \left(\frac{d}{2}\right)^2}} = \frac{0.3}{\sqrt{(0.308)^2 + (0.07)^2}} = 0.95$$

$$\cos \theta = \frac{\frac{d}{2}}{\sqrt{x^2 + \left(\frac{d}{2}\right)^2}} = \frac{0.07}{\sqrt{(0.3)^2 + (0.07)^2}} = 0.23$$

$$\vec{E}_+ = E_+ \sin \theta \hat{i} + E_+ \cos \theta (-\hat{j})$$

$$= \left(\frac{1}{4\pi\epsilon_0} \times \frac{|5|}{0.3^2 + 0.07^2} \right) \times 0.95 \hat{i}$$

$$+ \left(\frac{1}{4\pi\epsilon_0} \times \frac{|5|}{0.3^2 + 0.07^2} \right) \times 0.23 (-\hat{j})$$

$$= (4.5 \times 10^{11}) \hat{i} + (1.01 \times 10^{11}) (-\hat{j})$$

$$\vec{E}_- = E_- \sin \theta (-\hat{i}) + E_- \cos \theta (-\hat{j})$$

$$= \left(\frac{1}{4\pi\epsilon_0} \times \frac{|-4|}{0.3^2 + 0.07^2} \right) \times 0.95 (-\hat{i})$$

$$+ \left(\frac{1}{4\pi\epsilon_0} \times \frac{|-4|}{0.3^2 + 0.07^2} \right) \times 0.23 (-\hat{j})$$

$$= (3.6 \times 10^{11}) (-\hat{i}) + (8.7 \times 10^{10}) (-\hat{j})$$

$$\therefore \vec{E} = (4.5 \times 10^{11}) \hat{i} + (1.01 \times 10^{11}) (-\hat{j})$$

$$+ (3.6 \times 10^{11}) (-\hat{i}) + (8.7 \times 10^{10}) (-\hat{j})$$

$$= (9 \times 10^{10}) \hat{i} + (1.88 \times 10^{11}) (-\hat{j})$$

(Ans)

$$\therefore |\vec{E}| = \sqrt{(9 \times 10^{10})^2 + (1.88 \times 10^{11})^2} + 2 \cdot (9 \times 10^{10}) \cdot (1.88 \times 10^{11}) \cdot \cos 90^\circ$$

$$= 2.08 \times 10^{11} \text{ N/C}$$

(Ans)

$$b) \sin \theta = \frac{0}{\sqrt{0.07^2}} = 0$$

$$\cos \theta = \frac{0.07}{\sqrt{(0.07)^2}} = 1$$

$$\begin{aligned} \vec{E}_+ &= E_+ \sin \theta \hat{i} + E_+ \cos \theta (-\hat{j}) \\ &= 0 + \left(\frac{1}{4\pi\epsilon_0} \times \frac{151}{0^2 + 0.07^2} \right) (-\hat{j}) \\ &= 9.17 \times 10^{12} (-\hat{j}) \end{aligned}$$

$$\begin{aligned} \vec{E}_- &= E_- \sin \theta (-\hat{i}) + E_- \cos \theta (-\hat{j}) \\ &= 0 + \left(\frac{1}{4\pi\epsilon_0} \times \frac{(-9)}{0^2 + 0.07^2} \right) (-\hat{j}) \\ &= 7.34 \times 10^{12} (-\hat{j}) \end{aligned}$$

~~Ans)~~

$$\begin{aligned} \therefore \vec{E}_0 &= \vec{E}_+ + \vec{E}_- \\ &= 9.17 \times 10^{12} (-\hat{j}) + 7.34 \times 10^{12} (-\hat{j}) \\ &= 1.651 \times 10^{13} (-\hat{j}) \end{aligned}$$

(Ans)

c)

From b,

$$E = 1.651 \times 10^{13} (\hat{-j})$$

$$\vec{E} = \frac{\sigma}{2\epsilon_0}$$

$$\Rightarrow \vec{E} \times 2\epsilon_0 = \sigma$$

$$\Rightarrow \sigma = 1.651 \times 10^{13} \times 2\epsilon_0$$
$$= 292.37 \text{ C/m}^2$$

$$\therefore \sigma = 292.37 \text{ C/m}^2 \quad (\text{Ans})$$